



What Is a Variable?

Course: Intermediate Algebra · **Unit 1:** Expressions · **Deck:** U1Lo1_What_Is_a_Variable?.pptx (22 slides)

 You do not need to know algebra to teach this lesson. Read what is written. Follow the slides. Check answers against the key. The learning expectations for today are the same as any other day — this plan gives you the tools to keep the lesson moving. The art of teaching stays with your teacher; the tools are right here.

QUICK START — BEFORE THE BELL (2 MINUTES)

- ☐ Open the slide deck: **U1Lo1_What_Is_a_Variable?.pptx**
- ☐ Write the Bell Ringer on the board (see Slide 1 / Answer Key below)
- ☐ Test the projector
- ☐ Write your name on the board
- ☐ Find the answer key — it is printed inside this plan

MATERIALS FOR THIS LESSON

- ☐ Whiteboards, markers, erasers (1 set per student)
- ☐ Direct Instruction slide deck (Navy 1C2950 headings, Arial body)
- ☐ Tiered Independent Worksheet (T1/T2/T3 versions)
- ☐ Merit Time: Translation Match card sets (laminated, zero-device)
- ☐ Exit Ticket slips / LMS exit ticket
- ☐ Visual vocabulary anchor chart
- ☐ Theme visual: Minnesota State Fair

If anything is missing, check the teacher's desk or ask the front office.

PACING AT A GLANCE

Slide	By minute	Phase
1	0–4	Bell Ringer
7	15–16	End of Direct Instruction
19	42	End of Practice (worksheet + game/activity)
22	50	Exit Ticket & Wrap-Up

Pacing tip: bell ringer ~4 min · title/I-can/vocab ~4 min · collab ~4 min · worked example + guided practice ~7 min · culture + career ~6 min · GO/SET/EXTRA ~12 min · game/activity ~4 min · individual solve ~4 min · summary + exit ~4 min. Keep the timer visible; the deck drives the pace.

SUBSTITUTE SCRIPT (SLIDE BY SLIDE)

Slide 1 — Bell Ringer

On screen: WHITEBOARDS READY! — Tiered prompts: TIER 1 (Build Confidence) write an expression for basic situations with one operation · TIER 2 (Apply Your Skills) multi-step situations with two operations · TIER 3 (Challenge Yourself) complex situations with multiple operations. Caramel-apple expressions: $n + 3$, $2x - 5$, $3n + 4$, $4n - 7$, $2n + 3 = 15$.

Say: “Whiteboards ready — three tiers, four minutes. Choose your tier, work silently, show your steps, and hold your board up when the timer hits zero.”

Do: Walk the room as they work; cue “3... 2... 1 — boards up” and glance for accuracy.

Answer key: Tier 1: $n + 3$ (one operation). Tier 2: $2x - 5$ and $3n + 4$ (two operations). Tier 3: $4n - 7$ (multiple operations) and $2n + 3 = 15 \rightarrow$ subtract 3: $2n = 12 \rightarrow$ divide by 2: $n = 6$. Bridge: “The blank or letter stands for an unknown value — that letter is a variable.”

Slide 2 — Title

On screen: INTERMEDIATE ALGEBRA — UNIT 1: LESSON 1 — WHAT IS A VARIABLE?. Minnesota standards: 9.3.5.3, 9.3.5.7, 9.3.5.8. MERIT EDUCATION GROUP.

Say: “Welcome. Today’s lesson is Unit 1, Lesson 1: What Is a Variable. Everything you need is on these slides — follow along and write when I ask.”

Do: Read the lesson title aloud. Pause for students to find a pencil.

Slide 3 — I Can Statements

On screen: I CAN STATEMENTS — Define a variable; Write expressions from words; Evaluate expressions with one variable.

Say: “Here’s what we’ll be able to do by the end of the lesson. Read them silently; we’ll come back to these at the end.”

Do: Give 15 seconds, then continue.

Slide 4 — Vocabulary

On screen: VOCABULARY — Variable — a letter that stands for an unknown number (example: n); Coefficient — the number that multiplies a variable (example: $3n$); Expression — a combination of numbers, variables, and operations (example: $2n + 5$); Constant — a number that does not change (example: 7).

Say: “Quick vocabulary check. These are the words for today — I’ll read them, you repeat. [Read each term aloud.] We’ll use them all lesson.”

Do: Point to each term as you read it.

Slide 5 — Collaboration: Think-Pair-Share

On screen: ENGAGEMENT: THINK-PAIR-SHARE — prompt on the slide, tied to What Is a Variable?.

Say: “Think-Pair-Share. [Read the prompt aloud.] THINK silently for 30 seconds, then work with your partner or group, then we’ll share out.”

Do: Time 30s think, 45s pair, then take 2–3 volunteers.

Slide 6 — Worked Example (I Do)

On screen: WORKED EXAMPLE (I DO) — The number of books read, b ; twice as many books $\rightarrow 2b$. 1. IDENTIFY THE VARIABLE: let b represent the number of books read. 2. UNDERSTAND THE PHRASE: “twice as many books” means 2 times the number. 3. WRITE THE EXPRESSION: two times b is written $2b$. 4. FINAL EXPRESSION: $2b$.

Say: “A worked example — watch how it’s done step by step. The slide shows every step; I’ll read them as they appear.”

Do: Read each step aloud as it is revealed. Answer questions by pointing back to the slide.

Slide 7 — Guided Practice (We Do)

On screen: GUIDED PRACTICE (WE DO) — Context: pizza costs \$8 plus \$2 per topping. Let t represent the number of toppings. Expression: $8 + 2t$. Evaluate for $t = 3$: $8 + 2(3) = 8 + 6 = 14 \rightarrow \14 .

Say: “We do this one together. I’ll ask, you answer — then we check the slide.”

Do: Work step by step; call on students for the next step before revealing it.

Slide 8 — Culture

On screen: CULTURE: Ojibwe  (Tribal Nations). Ojibwe families harvest wild rice (manoomin) together from canoes, measuring and counting each harvest’s pounds of grain.

Say: “A cultural connection. [Read the caption aloud.] Notice how math shows up in everyday life and community traditions.”

Do: If the slide shows an image only, describe what students see; no math required.

Slide 9 — Culture Word Problem

On screen: CULTURE WORD PROBLEM — An Ojibwe elder counts $4x$ fish in one net and $3x$ fish in another net. Write an expression for the total number of fish. If $x = 7$ fish per net, how many fish are there total?

Say: “Here’s the cultural word problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide: circle, underline, box, write.”

Do: Give 30 seconds to read; walk through the strategy, then the solution.

Slide 10 — Career

On screen: CAREER: MEDICAL LAB SCIENTIST — Mayo Clinic, Rochester, MN. Entry wage: \$63,150/year (MN DEED OES, 25th percentile). Median wage: \$78,458/year.

Say: “A real career uses this math every day. [Read the career spotlight aloud.] The math we’re learning today is the math people use at work.”

Do: Read confidently; the salary and role are on the slide.

Slide 11 — Career Word Problem

On screen: CAREER WORD PROBLEM: COMMUNITY GARDEN — \$12 registration fee plus \$4 per plot. For p plots, total cost: $\$12 + \$4p$. Evaluate: if $p = 6$, $\$12 + \$4(6) = \$12 + \$24 = \$36$. Sarah Chen, Medical Lab Scientist at Mayo Clinic and community volunteer. SOLVE LIKE A PRO! Reading strategy: circle, underline, box, write.

Say: “Here’s the career problem. Read it to yourself — we’ll solve it together later in the lesson.”

Do: Give 30 seconds. Walk the reading strategy; then the solution.

Answer key: $\$12 + \$4p$; when $p = 6 \rightarrow \$12 + \$24 = \$36$.

Slide 12 — GO Practice (Your Turn)

On screen: GO PRACTICE: YOUR TURN — A number n is tripled then increased by 4. Write an algebraic expression

Say: “Your turn — a problem just like the one we did. Work it on your whiteboard (or paper). Two minutes.”

Do: Circulate. Cue boards up at 2 minutes.

Answer key: $3n + 4$

Slide 13 — GO Answer

On screen: GO ANSWER — reveal: $3n + 4$.

Say: “Let’s check it. The slide walks through the solution step by step — I’ll read each step. Check your work against it; fix anything you missed.”

Do: Read the solution steps aloud.

Slide 14 — SET Practice

On screen: SET PRACTICE — two problems: x decreased by 9. The product of 6 and a number y .

Say: “Two practice problems to deepen your understanding. Work them on paper or whiteboard. You have 5 minutes; I’ll be walking around.”

Do: Circulate and support. Cue time at 3 and 1 minutes.

Slide 15 — SET Answer

On screen: SET ANSWER — reveal: $x - 9$ and $6y$.

Say: “Let’s check. Read the steps on the slide; self-correct with a pen.”

Do: Read the steps aloud; answer questions by re-reading the relevant step.

Slide 16 — EXTRA Practice Challenge

On screen: EXTRA PRACTICE CHALLENGE — Twice a number increased by 3 equals 15. Write an equation and find the number

Say: “Challenge problem — for anyone who wants a stretch. Try it on your own paper. We’ll go over it in a moment.”

Do: Encourage all to attempt; it is optional.

Slide 17 — EXTRA Answer

On screen: EXTRA ANSWER — reveal: $2n + 3 = 15$, $n = 6$.

Say: “Let’s check the challenge. [Read the steps.] Great work.”

Do: Read the solution steps aloud; celebrate attempts.

Slide 18 — Merit Time Game: Translation Match

On screen: MERIT TIME GAME: TRANSLATION MATCH — SETUP: 4–6 teams, each team gets 16 cards (8 prompts + 8 matches). PLAY: teams race to complete the match. First team with all 8 correct wins! SCORING: 1 point per correct match + 2 bonus points for the first team finished. ZERO-DEVICE · WHOLE-CLASS.

Say: “Quick whole-class game — we do this as a team, quietly. I’ll read the rules from the slide.”

Do: Read the rules; run the game as the slide directs; keep it positive.

Game from lesson plan: Merit Time: Translation Match (zero-device). Students work in teams to match prompts to answers. Teacher facilitates verification and corrects common errors aloud.

Slide 19 — Individual Solve: Tiered Worksheet

On screen: INDIVIDUAL SOLVE: TIERED WORKSHEET — TIER 1 Build Confidence (6 questions), TIER 2 Apply Your Skills (6 questions), TIER 3 Challenge Yourself (6 questions). PROBLEMS: TIERED WORKSHEET — WHAT IS A VARIABLE? + IN YOUR LMS. SHOW YOUR WORK.

Say: “Independent work now — the tiered worksheet. Pick your tier, show your work, and submit in the LMS when you finish.”

Do: Circulate; support tier choices; collect or confirm LMS submission.

Slide 20 — Summary

On screen: SUMMARY — RECAP: key ideas from What Is a Variable?. REVIEW: the skills practiced today. [Slide lists the takeaways.]

Say: “Three big takeaways from today. [Read them aloud.] This is what you should remember tomorrow.”

Do: Read each takeaway; pause between them.

IF A STUDENT ASKS...

“Why are we doing this?” — “This is today’s learning expectation for What Is a Variable?. It builds toward what’s next in the unit, and your teacher will pick up right here tomorrow.”

Redirects (from the lesson plan):

Misconception	Redirect Question	Corrective Strategy
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| Confuses variable with constant | “Which part of the expression changes depending on the situation?” | Use color-coding: variables in blue, constants in black. |

| Writes $5 \times n$ instead of $5n$ | “How do mathematicians abbreviate multiplication between a number and a letter?” | Model implicit multiplication; emphasize coefficient placement. |

| Translates “3 less than x” as $3 - x$ | “Read the phrase carefully: what is being decreased, and by how much?” | Use arrow diagrams or sentence frames to reverse the order. |

IF A STUDENT FINISHES EARLY

- Students who finish early do the **Challenge** side of the worksheet — do not invent extra work.
- If the whole class is ahead: re-read the Summary slide and ask students to explain one takeaway in their own words.
- **Do not** move ahead in the curriculum.

IF THE PROJECTOR DIES

Write the Bell Ringer problems on the board (from the Answer Key).

Read the worksheet directions aloud; students work in pairs.

Use the printed answer key to check as a class.

The lesson still runs — the script lives in this plan, not the projector.

WITH A PARAPROFESSIONAL OR CO-TEACHER

If an adult is in the room supporting students, they know the students best — **follow their lead** on individual support. Check the roster for accommodations; extended time on the Exit Ticket is fine if indicated.

SUB REPORT (LEAVE FOR THE TEACHER)

What did we complete? (bell ringer / DI / practice / game / exit ticket)

Who struggled or needed extra support?

Behavior notes (positive first, then concerns)

Anything tech-related that went wrong

TEACHER'S OWN NOTES

(The regular teacher may add specifics here — their timer, signals, roster quirks.)

Generated by the MEG Substitute Plan pipeline from the Lesson Plan + Teacher Edition — synced to the published slide decks (2026-08-19).